MODULE 19

Surveillance and CCTV Investigation

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

No eyewitness, no clues - but a shop camera saw everything. **CCTV is the investigator's most trusted eye.**

- How CCTV, **DVR and NVR** work - and where footage actually lives
- **Forensic imaging** from a DVR and native-format export
- Proving authenticity with **hash values** - certutil and sha256sum
- Spotting **tampered footage** - frame by frame
- Seizure memos, chain of custody, and **court presentation**

The Bike Chased Camera to Camera

A theft. No witness. But a corner shop's camera caught a **suspicious bike** - then the next crossing's camera, then the next.

- Camera by camera, the police **followed the bike to the thief**
- This is not a movie - it is **real daily investigation**
- This module teaches you to **seize, verify and read** what the camera saw
- One rule first: DVRs **overwrite in 7 to 15 days** - delay is the biggest mistake

Collecting CCTV Evidence

Golden rule: preserve the footage in its **original form** - a bit-for-bit copy. No cutting, no compression.

- First find the cameras: government, shop, home, ATM - **ask on day one**
- Write a **seizure memo**: when, where, from which device, who was present
- Note the **hash value** - the digital fingerprint that proves genuineness later
- Remember **overwriting**: full disks auto-delete the oldest recording - securing footage is the **first action**

Cameras and DVRs: The Basics

Two camera types, two recorders - **know what you are seizing**:

- **Analog cameras:** older tech, coaxial cable into a DVR; **IP cameras:** network cable or WiFi into an NVR - sharper, modern
- **Resolution** (720p, 1080p, 4MP) decides detail; **FPS** (frames per second) decides how smooth movement looks
- **DVR** = Digital Video Recorder; **NVR** = Network Video Recorder for IP cameras
- Quality decides everything downstream - **note the specs in your memo**

Native Format and the Time Offset

DVRs speak their own language - and keep their own time. **Both facts shape your evidence.**

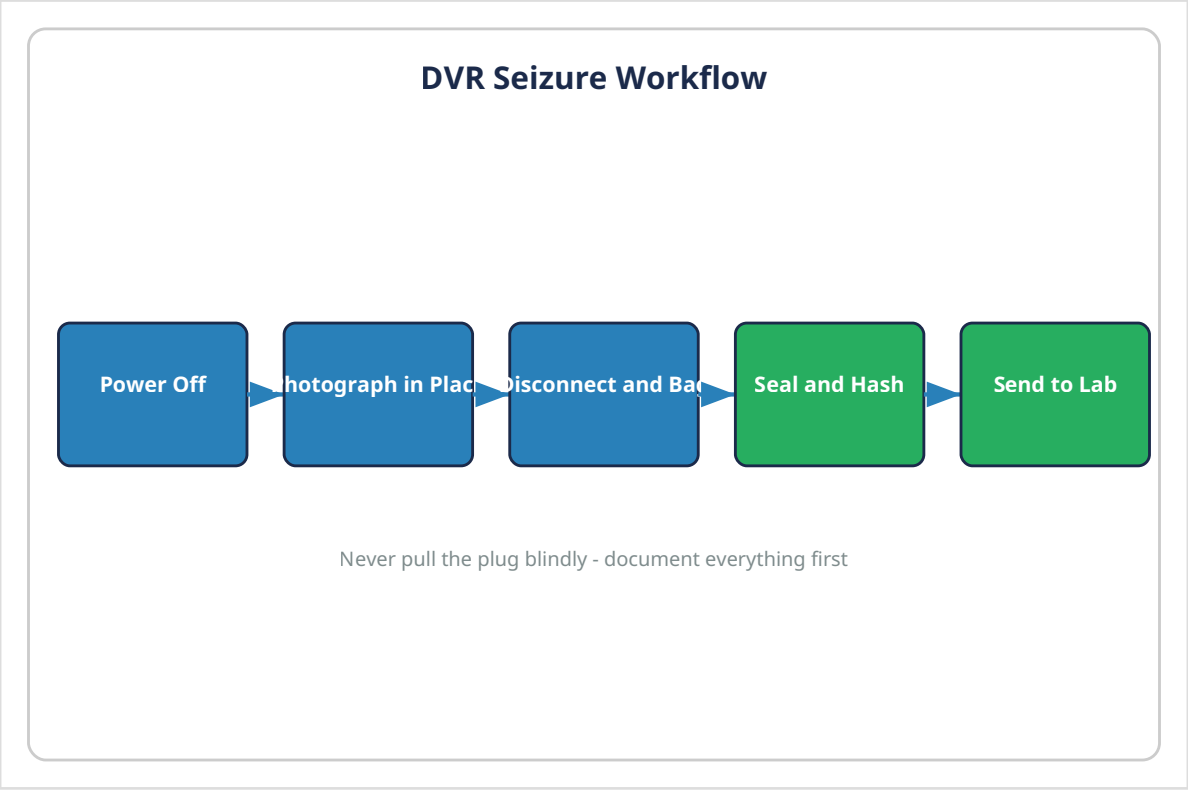
- Many DVRs record in **DAV or H264** - unplayable in common players
- Make **two copies**: native (as the DVR gave it - the real evidence) and common MP4/AVI (for viewing)
- **Time offset:** DVR clocks are often wrong - note DVR time **and** your phone's real time together
- A 12-minute offset, uncorrected, **breaks your whole timeline**

Forensic Imaging: The Exact Copy

Extracting from a DVR is delicate - **the method is the evidence**:

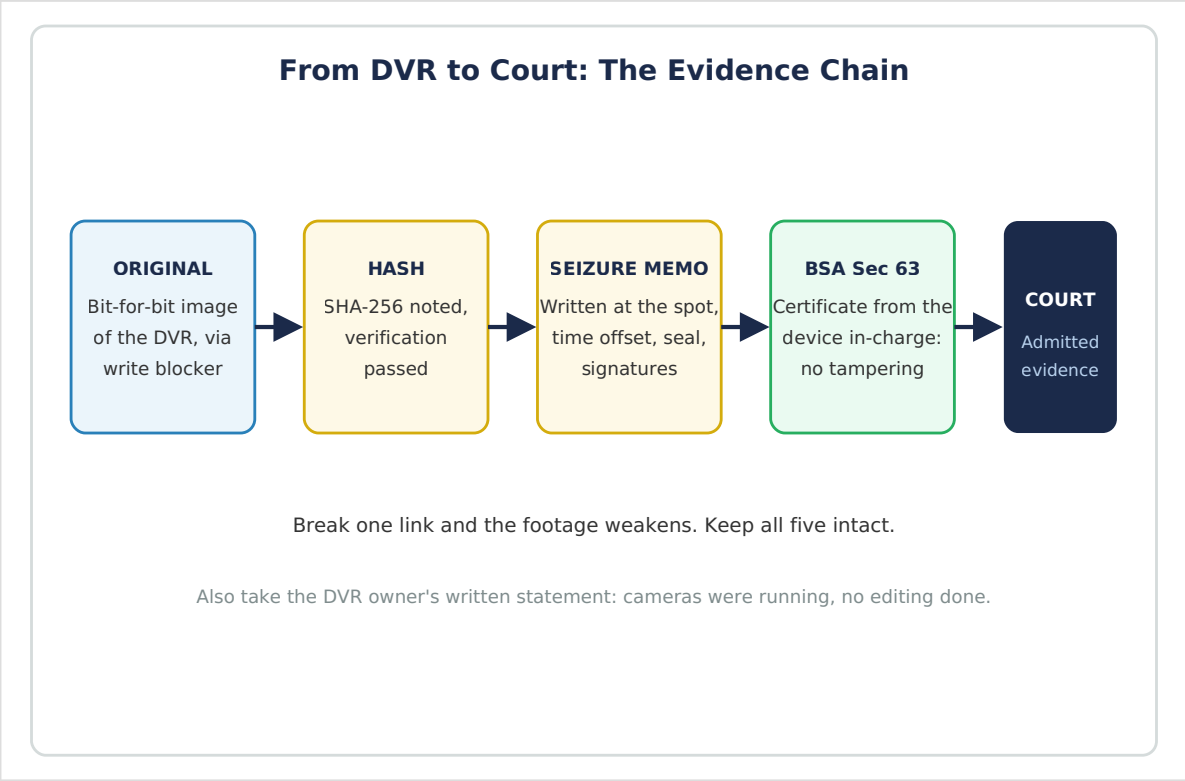
- **Forensic image:** a bit-for-bit copy of the entire DVR, made without changing the original
- **Do not switch it off** without expert advice - wrong shutdown corrupts recordings; image it running
- Use a **write blocker** - reads allowed, writes blocked - the first link of chain of custody
- Take the clip from **before and after** the crime window - arrival and escape routes matter as much as the act

DVR Seizure Workflow



- Five steps: power decision, photograph, disconnect, seal, hash
- **Photograph first** - cabling, screen, condition
- Every step **witnessed and written**

From DVR to Court



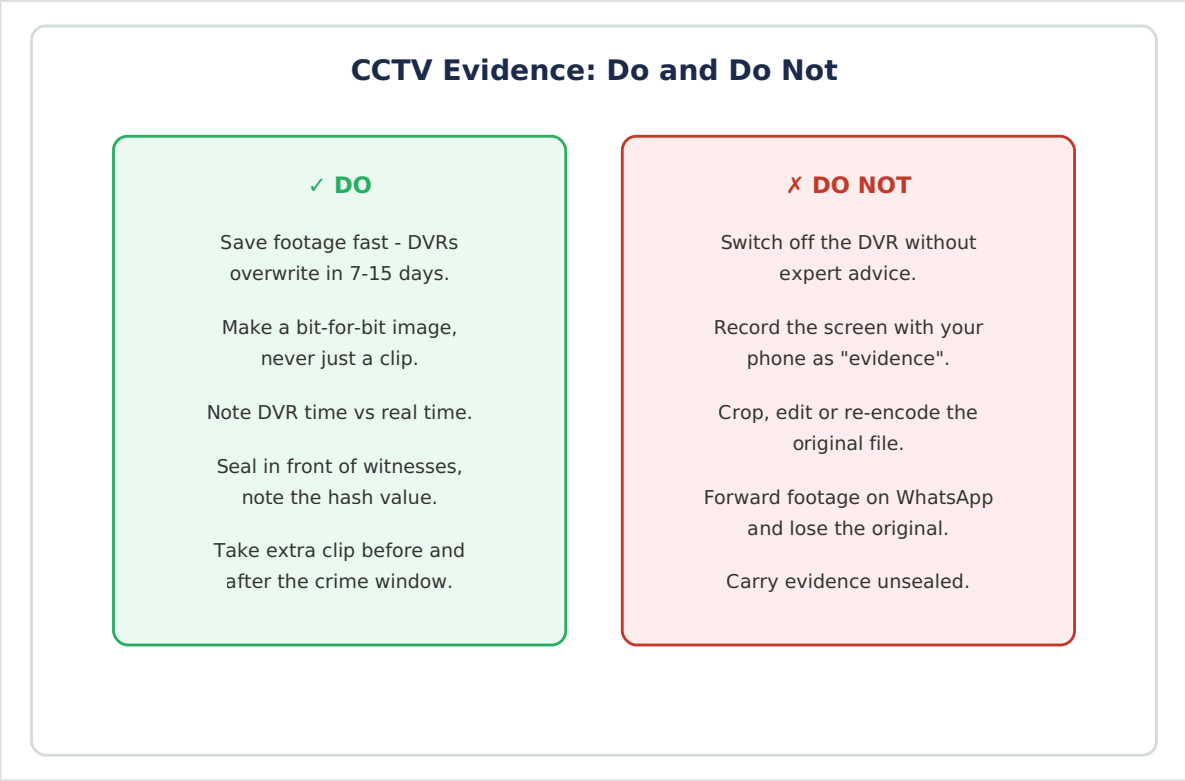
- Five links: original, hash, memo, BSA s.63, court
- Break one link and the **footage weakens**
- Also take the owner's **written statement**

The Seizure Toolkit

Carry it ready - **CCTV calls come without warning**:

- **Write blocker** - the single most important tool; evidence stays exactly as it was
- **Forensic laptop** plus a high-capacity **empty disk** for the image; FTK Imager installed
- Screwdriver set, **camera** for the scene, printed **seizure memo forms**
- **Seal and label everything** - unsealed evidence is weak evidence in court

CCTV Evidence: Do and Do Not



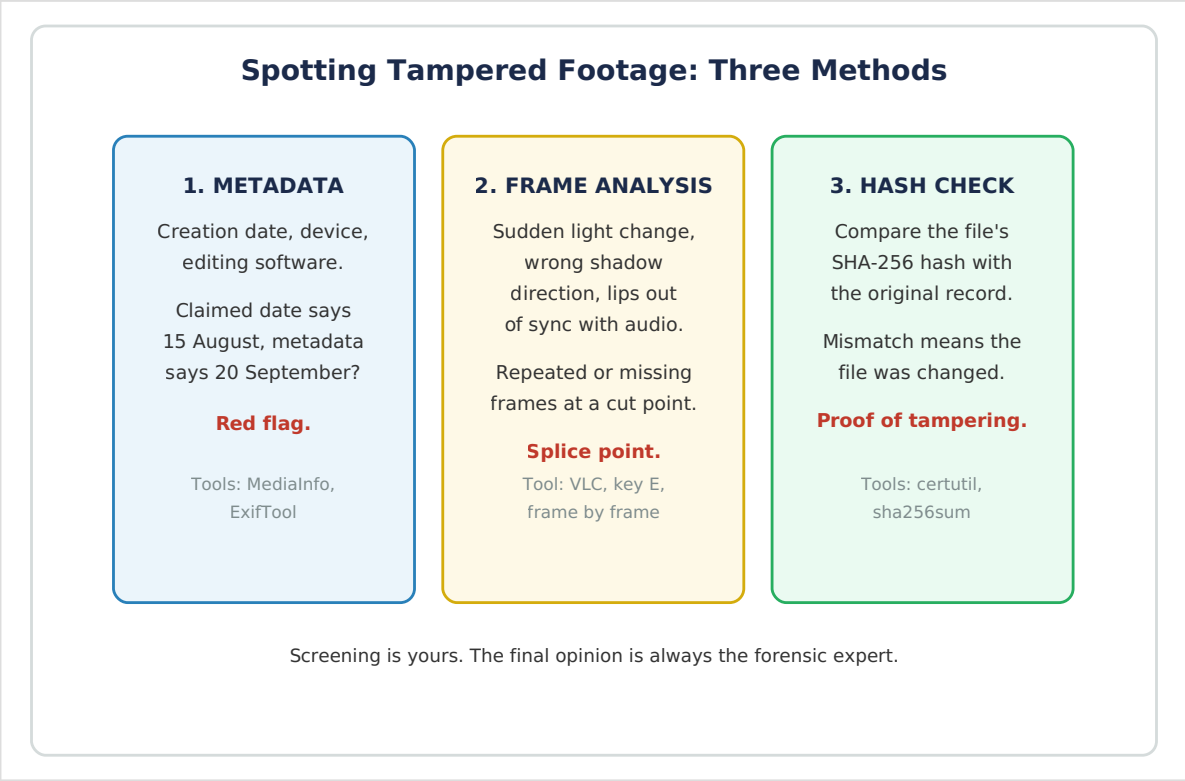
- Ten rules - **five gold, five fatal**
- One mistake can **weaken the entire case**
- Only correctly collected footage **holds in court**

Spotting Tampered Footage

Cut a part? Changed a face? Faked a date? **The file usually confesses** - if you know where to look:

- **Metadata:** creation date, device, editing software - claimed 15 August but file says 20 September?
- **Frames:** sudden light shifts, wrong shadow direction, lips out of sync, repeated or missing frames
- **Hash:** mismatch with the original record means the file **was changed**
- Screen it yourself - but the **final opinion is the forensic expert's**, never yours in court

Three Tamper-Detection Methods



- Metadata, frames, hash - **in that order**
- Free tools: MediaInfo, ExifTool, VLC
- Write **observations**, not opinions

Frame by Frame in Practice

VLC plus patience: **press E** and walk through time one frame at a time.

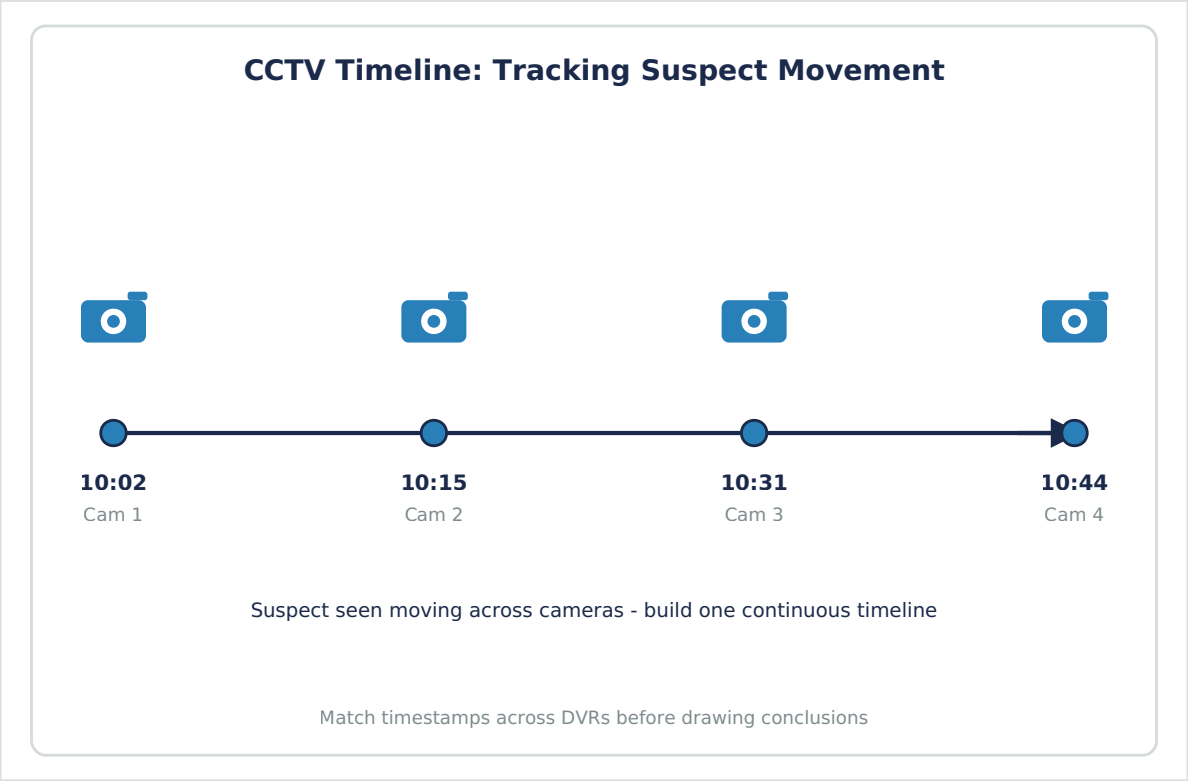
- Note **codec info** first: resolution, frame rate, video codec - your baseline
- At the suspicious point: **jump, repeat, or missing frame** means a possible splice
- Check **audio sync** and background noise shifts; zoom into compression artifacts at edges
- Log every suspicion with a **timestamp** - 00:02:14 light shift, 00:03:40 audio cut - then hand to the expert

Building the Multi-Camera Timeline

One camera is a clip. **Many cameras are a story** - joined by corrected time:

- Camera 1: arrival. Camera 2: the route. Camera 3: the act. **Join by real time**, not DVR time
- Correct **every camera's offset** - one runs 12 minutes fast, another 5 slow
- Each entry: camera name, DVR time, **real time**, event seen
- **Enhance is not invent:** brighten, deblur - but what is not in the footage cannot be added

The CCTV Timeline



- Many cameras, **one corrected timeline**
- Real time on every entry - **no exceptions**
- Judges **understand timelines**

Presenting Footage in Court

Electronic evidence needs a **certificate** - arrange it at seizure time, not trial time:

- **BSA Section 63:** certificate from the device in-charge - footage from the original device, unaltered
- Without it, the defence **questions everything** - get it the same day
- Owner's written statement: cameras were running, DVR worked properly, **no editing done**
- Present the **chain of custody** alongside - hash, memo, custody log: the cleaner the record, the stronger the footage

Citizen Evidence Counts Too

Today **everyone carries a camera**. A bystander's phone video can crack your case:

- Same rules apply: **genuine, untampered**, maker's statement recorded, phone examinable
- Social media videos: **download immediately** - the uploader can delete anytime
- Lesson: evidence is not only CCTV - **every digital trace matters**
- Handle a citizen's phone with the **same care** as a DVR

Legal Surveillance: Know Your Limits

The camera never blinks - but **the law watches the watcher**:

- Phone tapping, location tracking, secret surveillance need **legal process** - usually senior permission, often a court order
- **Public-place CCTV** is legal surveillance; aiming a camera **inside someone's home** can violate privacy
- Never **break the law while collecting evidence** - tainted collection poisons the case
- A good investigator catches the criminal **and** stays inside the law

Mistakes That Kill Footage Value

Five ways good footage becomes **useless evidence**:

- **Delay:** the DVR overwrote the crime window while you finished paperwork
- **Phone-recording the screen** as 'evidence' - quality and authenticity both questioned
- **Editing the original** - crop, re-encode, or 'clean up' destroys the hash match
- **WhatsApp forwarding** and losing the original - compression kills metadata
- Carrying it **unsealed** - the defence will ask who else touched it

Real Case: The Tempo Chased Through 50 Cameras

Delhi. A minor's body found. **No eyewitness, no nearby camera** - only a tempo glimpsed at a distance, plate unreadable in the dark.

- One clue: a **strange round mark and an AC unit** on the tempo's left side
- Police traced it through **about 50 CCTV cameras** - an all-night operation, camera by camera
- Result: tempo found, **four caught**, knives, clothes and the tempo recovered
- Source News18
- Lesson: **patience is the weapon** - one small mark, hours of footage, justice

Real Case: 3.7 Crore Gold Recovered in 24 Hours

Nadiad railway station, Gujarat. A passenger's bag stolen on the foot overbridge - **2.7 kg of gold inside**.

- RPF watched station and nearby cameras - **movements and escape route** became clear
- Footage shared with GRP; **next day, all four arrested**, all property recovered - a former railway employee among them
- In 2026 alone: **55 luggage-theft cases** solved via CCTV, 68 accused, 3.83 crore property recovered
- Source The Indian Express

Your Turn: Labs in This Module

Four labs - **do them in order**, on your own devices:

- **Lab 1:** Extract DVR footage, image it, record SHA-256 with certutil or sha256sum
- **Lab 2:** Frame-by-frame tamper screening with VLC, MediaInfo and ExifTool
- **Lab 3:** Write a complete seizure memo - device, hashes, offset, seal, signatures
- **Lab 4:** Map cameras on a 200-metre stretch - blind spots, offsets, one multi-camera timeline

Key Takeaways

If you remember nothing else from Module 19:

- **Original form, fast** - bit-for-bit copy before the DVR overwrites
- **Write blocker always** - the first link of chain of custody
- **Hash plus memo plus BSA s.63** - the trio that makes footage court-strong
- Correct **every camera's time** - timelines live or die on offsets
- Screen for tampering; let the **expert conclude**
- Stay **inside the law** while watching - the watcher is watched too

Review Questions

- 1. Why is a write blocker used? Describe exactly what goes wrong without one.
- 2. Write the full Windows command for a SHA-256 hash. What does a mismatch prove?
- 3. Name three splice-point signs you can find with VLC alone.
- 4. A DVR shows 14:47, your phone shows 14:35. A crime is visible at DVR time 14:20. What is the real time - and why does the memo need both?
- 5. Two cameras, one 10 minutes fast, one 5 slow. Explain step by step how you build one correct timeline.

NEXT: MODULE 20

GPS, Cell IDs and IMEI

Your phone is a location diary - learn to read it lawfully

Cyber Crime Investigator - Training Manual